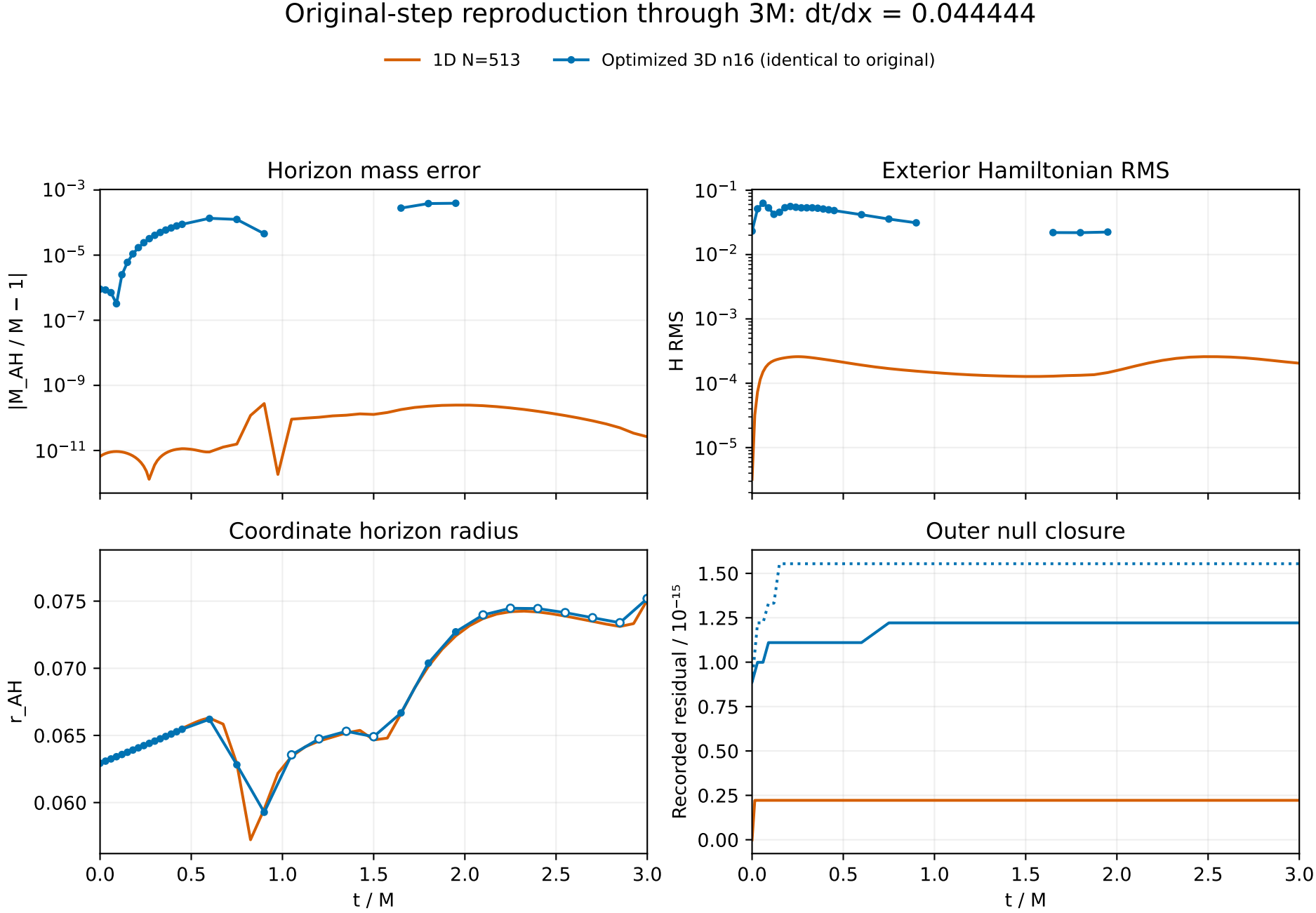

Open radius markers are complete horizon candidates that miss the residual tolerance. Missing norms are not filled.
1D is 5.76× finer than the 3D core. Dotted blue line: ghost null closure. This does not establish convergence.